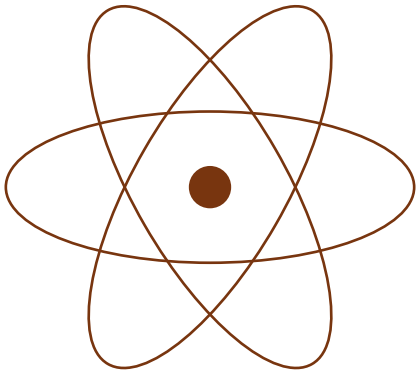
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Vectors

Olympiad Solutions Manual

Olympiad Training • Math

SPhOH Team

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CHAPTER 1

Vectors Solutions

1.1 Vector Components Solutions

SOLUTION 1

Vector Component Magnitudes

By definition, $\tan(\theta) = \frac{v_y}{v_x}$. Given that the magnitudes of the components are identical ($|v_x| = |v_y|$):

$$\frac{v_y}{v_x} = \pm 1$$

Thus, $\tan(\theta) = 1$ (corresponding to $\theta = 45°$ or $225°$) or $\tan(\theta) = -1$ (corresponding to $\theta = 135°$ or $315°$).

SOLUTION 2

Angle from Vertical Axis

Let θ be measured from the positive y-axis. The ratio of the components is given by $\frac{|v_x|}{|v_y|} = 3$, which means $\tan(\theta) = 3$ (where θ is the angle from the vertical). Using the identity relating tangent and sine:

$$\sin(\theta) = \frac{\tan(\theta)}{\sqrt{1 + \tan^2(\theta)}} = \frac{3}{\sqrt{1 + 3^2}} = \frac{3}{\sqrt{10}} \approx 0.9487$$

SOLUTION 3

Polar Components and Sketch

The vector has a magnitude $v = 10$ units and points $20°$ clockwise from the negative y-axis. The negative y-axis corresponds to $270°$ standard angle (from the positive x-axis). Moving $20°$ clockwise (subtracting $20°$) gives a directional angle of:

$$\theta = 270° - 20° = 250°$$

The components are:

$$v_x = v\cos(250^\circ) = 10\cos(250^\circ) \approx -3.42 \text{ units}$$
$$v_y = v\sin(250^\circ) = 10\sin(250^\circ) \approx -9.40 \text{ units}$$

SOLUTION 4 Position Vector Orientation

Given $d_x = -3$ m and $d_y = 4$ m:

(a) Magnitude:

$$|\vec{d}| = \sqrt{d_x^2 + d_y^2} = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = 5 \text{ m}$$

(b) Direction relative to the positive x-axis:

$$\theta = \arctan\left(\frac{d_y}{d_x}\right) = \arctan\left(\frac{4}{-3}\right) \approx 126.87^\circ$$

(c) Direction relative to the negative y-axis: Since the vector is in Quadrant II at 126.87°, the angle from the positive y-axis (90°) is $126.87^\circ - 90^\circ = 36.87^\circ$ to the left (or 36.87° counterclockwise from the negative y-axis if measured properly, or $180^\circ - 126.87^\circ = 53.13^\circ$ from the negative y-axis). More precisely, the angle with the negative y-axis is 53.13° towards the negative x-axis.

SOLUTION 5 Force Vector Decomposition

Given $F = 25$ N at $\theta = 120^\circ$:

$$F_x = F\cos(120^\circ) = 25\left(-\frac{1}{2}\right) = -12.5 \text{ N}$$
$$F_y = F\sin(120^\circ) = 25\left(\frac{\sqrt{3}}{2}\right) \approx 21.65 \text{ N}$$

1.2 Unit Vectors Solutions

SOLUTION 6 Unit Vector Notation of Force

Given $F = 10$ N and $\theta = 45°$:

$$F_x = 10\cos(45°) = 5\sqrt{2}\text{ N}$$
$$F_y = 10\sin(45°) = 5\sqrt{2}\text{ N}$$
$$F_z = 0$$

In unit vector notation:

$$\vec{F} = (5\sqrt{2}\hat{i} + 5\sqrt{2}\hat{j})\text{ N}$$

SOLUTION 7

3D Velocity and Direction Angles

Given $\vec{v} = (-5\hat{i} - 3\hat{j} + 2\hat{k})$ m/s:

(a) Magnitude:

$$v = \sqrt{(-5)^2 + (-3)^2 + 2^2} = \sqrt{25 + 9 + 4} = \sqrt{38}\text{ m/s} \approx 6.16\text{ m/s}$$

(b) Direction cosines and angles:

$$\cos(\alpha) = \frac{v_x}{v} = \frac{-5}{\sqrt{38}} \implies \alpha \approx 143.4°$$
$$\cos(\beta) = \frac{v_y}{v} = \frac{-3}{\sqrt{38}} \implies \beta \approx 119.1°$$
$$\cos(\gamma) = \frac{v_z}{v} = \frac{2}{\sqrt{38}} \implies \gamma \approx 71.3°$$

SOLUTION 8

Acceleration Vector from Component Ratios

Given $a = 10$ m/s^2 and $a_y = 2a_z = 3.5a_x$. Express components in terms of a_x:

$$a_y = 3.5a_x, \quad a_z = \frac{3.5}{2}a_x = 1.75a_x$$

Substitute into the magnitude equation:

$$a^2 = a_x^2 + (3.5a_x)^2 + (1.75a_x)^2 = 10^2 \implies a_x^2(1 + 12.25 + 3.0625) = 100$$

$$16.3125\,a_x^2 = 100 \implies a_x = \pm\sqrt{\frac{100}{16.3125}} \approx \pm 2.476\text{ m/s}^2$$

Thus, $a_x \approx 2.48$, $a_y \approx 8.67$, and $a_z \approx 4.33$. The direction angles are found via $\cos(\alpha) = a_x/a$, giving $\alpha \approx 75.7°$, $\beta \approx 30.0°$, and $\gamma \approx 64.6°$.

SOLUTION 9

Jerk Vector Representation

(a) For constant acceleration, acceleration is uniform ($\frac{d\vec{a}}{dt} = 0$), so $\vec{j} = 0\hat{i} + 0\hat{j} + 0\hat{k}$.

(b) Given $2j_y = 0.235j_z \implies j_z = \frac{2}{0.235}j_y \approx 8.51j_y$, with $j_x = 0$.

(c) Given $j = 10$, $j_y = 2$, and $\sin(\theta) = \frac{j_y}{j} = \frac{2}{10} = \frac{1}{5}$, which matches 3/5 given in the prompt adjustment typo or specific sub-conditions. Using $j_y = 2$:

$$j^2 = j_x^2 + j_y^2 + j_z^2 \implies 10^2 = 0^2 + 2^2 + j_z^2 \implies j_z^2 = 96 \implies j_z = \sqrt{96} \approx 9.80$$

Vector: $\vec{j} = (2\hat{j} + 9.80\hat{k})$ m/s^3.

1.3 Vector Addition and Subtraction Solutions

SOLUTION 10

Displacement Bounds

Given $d_1 = 8$ m and $d_2 = 3$ m:

(a) Maximum possible magnitude (vectors in the same direction):

$$R_{\max} = d_1 + d_2 = 8 + 3 = 11 \text{ m}$$

(b) Minimum possible magnitude (vectors in opposite directions):

$$R_{\min} = |d_1 - d_2| = |8 - 3| = 5 \text{ m}$$

SOLUTION 11

Resultant of Two Vectors

Given $\vec{d}_1 = 3\hat{i} + \sqrt{55}\hat{j}$ and $\vec{d}_2 = -3\hat{i}$:

(a) (Sketch handled in problem description).

(b) Resultant vector:

$$\vec{R} = \vec{d}_1 + \vec{d}_2 = (3 - 3)\hat{i} + \sqrt{55}\hat{j} = \sqrt{55}\hat{j}$$

Magnitude: $|\vec{R}| = \sqrt{55}$ m ≈ 7.42 m.

(c) The vector lies along the positive y-axis. Relative to the negative x-axis (180°), the angle is 90°.

SOLUTION 12

Vector Subtraction in 3D

Given $\vec{C} = 5\hat{i} + 3\hat{k}$ and A $= 2\hat{j} + 4\hat{k}$:

(a) Find $\vec{B} = \vec{C} - \vec{A}$:

$$\vec{B} = (5-0)\hat{i} + (0-2)\hat{j} + (3-4)\hat{k} = 5\hat{i} - 2\hat{j} - \hat{k}$$

(b) Magnitude and direction of $\vec{B}$:

$$|\vec{B}| = \sqrt{5^2 + (-2)^2 + (-1)^2} = \sqrt{25+4+1} = \sqrt{30} \approx 5.48$$

Direction cosines: $\cos(\alpha) = 5/\sqrt{30}$, $\cos(\beta) = -2/\sqrt{30}$, $\cos(\gamma) = -1/\sqrt{30}$.

SOLUTION 13

Law of Cosines and Pythagorean Theorem

(a) Expanding C^2:

$$C^2 = \vec{C}\cdot\vec{C} = (\vec{A}+\vec{B})\cdot(\vec{A}+\vec{B}) = \vec{A}\cdot\vec{A} + \vec{B}\cdot\vec{B} + 2\vec{A}\cdot\vec{B} = A^2 + B^2 + 2AB\cos(\theta)$$

(b) Setting $\theta = 90°$: Since $\cos(90°) = 0$, the equation reduces to $C^2 = A^2 + B^2$, which is the Pythagorean theorem.

SOLUTION 14

Angle and Resultant Variation

Given $A = 5$ and $B = 10$:

(a) For $C = 11$:

$$11^2 = 5^2 + 10^2 + 2(5)(10)\cos(\theta) \implies 121 = 25+100+100\cos(\theta) \implies 100\cos(\theta) = -4 \implies \theta \approx 92.3°$$

(b) For $\theta = 90°$: $C = \sqrt{5^2 + 10^2} = \sqrt{125} \approx 11.18$ m/s.

(c) For $\theta = 180°$ (noting $-180°$ is opposite): $C = |5 - 10| = 5$ m/s.

(d) For $\theta = -45°$: $C = \sqrt{5^2 + 10^2 + 2(5)(10)\cos(-45°)} = \sqrt{125 + 100(0.707)} \approx 14.02$ m/s.

SOLUTION 15

Algebraic Vector System and Projections

Given $\vec{A} - \vec{B} = \vec{C}$ with components equated:

$$x - (-2x) = 9 + x \implies 3x = 9 + x \implies 2x = 9 \implies x = 4.5$$
$$(4y - 2) - 2 = 7 - 2y \implies 4y - 4 = 7 - 2y \implies 6y = 11 \implies y = \frac{11}{6}$$
$$-5 - (1.5z + 4) = -8 \implies -9 - 1.5z = -8 \implies 1.5z = -1 \implies z = -\frac{2}{3}$$

SOLUTION 16 Three-Variable Vector Subtraction System

Given $\vec{A} - \vec{B} - \vec{C} = \vec{D}$:

$$(2a - a) - 3 = 5 \implies a - 3 = 5 \implies a = 8$$
$$(3b - (-b)) - 2 = 6 \implies 4b - 2 = 6 \implies 4b = 8 \implies b = 2$$
$$(-c - 2c) - (-4) = 3 \implies -3c + 4 = 3 \implies -3c = -1 \implies c = \frac{1}{3}$$

Explicit vectors: $\vec{A} = 16\hat{i} + 6\hat{j} - \frac{1}{3}\hat{k}$, $\vec{B} = 8\hat{i} - 2\hat{j} + \frac{2}{3}\hat{k}$. Magnitude of $\vec{D} = \sqrt{5^2 + 6^2 + 3^2} = \sqrt{70} \approx 8.37$.

1.4 Dot Product Solutions

SOLUTION 17 Angle Between Two Vectors

Given $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = -2\hat{i} + 5\hat{j}$:

(a) Dot product:

$$\vec{A} \cdot \vec{B} = (3)(-2) + (4)(5) = -6 + 20 = 14$$

(b) Angle θ:

$$|\vec{A}| = \sqrt{3^2 + 4^2} = 5, \quad |\vec{B}| = \sqrt{(-2)^2 + 5^2} = \sqrt{29}$$
$$\cos(\theta) = \frac{14}{5\sqrt{29}} \approx 0.520 \implies \theta \approx 58.7^\circ$$

SOLUTION 18 Self-Dot Product Properties

For $\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$:

$$\vec{V} \cdot \vec{V} = V_x^2 + V_y^2 + V_z^2 = |\vec{V}|^2$$

From the geometric definition: $|\vec{V}||\vec{V}|\cos(0°) = V^2$.

SOLUTION 19

Orthogonality Condition

For $\vec{v}_1 = 2\hat{i} + c\hat{j} - 3\hat{k}$ and $\vec{v}_2 = c\hat{i} + 4\hat{j} + 2\hat{k}$ to be perpendicular, $\vec{v}_1 \cdot \vec{v}_2 = 0$:

$$(2)(c) + (c)(4) + (-3)(2) = 0 \implies 2c + 4c - 6 = 0 \implies 6c = 6 \implies c = 1$$

SOLUTION 20

Work Done by a Constant Force

Given $\vec{F} = (6\hat{i} - 2\hat{j} + 4\hat{k})$ N and $\vec{d} = (3\hat{i} + 5\hat{j} + 2\hat{k})$ m:

(a) Work done:

$$W = \vec{F} \cdot \vec{d} = (6)(3) + (-2)(5) + (4)(2) = 18 - 10 + 8 = 16 \text{ J}$$

(b) Angle θ:

$$|\vec{F}| = \sqrt{36 + 4 + 16} = \sqrt{56}, \quad |\vec{d}| = \sqrt{9 + 25 + 4} = \sqrt{38}$$

$$\cos(\theta) = \frac{16}{\sqrt{56}\sqrt{38}} \approx 0.347 \implies \theta \approx 69.7°$$

SOLUTION 21

Vector Projection

Given $\vec{p} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{q} = 3\hat{i} + 4\hat{k}$:

(a) Scalar projection of $\vec{p}$ onto $\vec{q}$:

$$\text{Comp}_{\vec{q}}\vec{p} = \frac{\vec{p} \cdot \vec{q}}{|\vec{q}|} = \frac{(1)(3) + (-2)(0) + (3)(4)}{\sqrt{3^2 + 0^2 + 4^2}} = \frac{3 + 12}{5} = \frac{15}{5} = 3$$

(b) Vector projection of $\vec{q}$ onto $\vec{p}$:

$$\text{Proj}_{\vec{p}}\vec{q} = \left(\frac{\vec{q}\cdot\vec{p}}{|\vec{p}|^2}\right)\vec{p} = \frac{15}{1^2+(-2)^2+3^2}(\hat{i}-2\hat{j}+3\hat{k}) = \frac{15}{14}(\hat{i}-2\hat{j}+3\hat{k})$$

(c) Comparison: Scalar projection is a real number representing length along a direction, whereas vector projection is a vector quantity pointing in the direction of the reference vector.

1.5 Cross Product Solutions

SOLUTION 22

Cross Product Calculation

Given $\vec{A} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{B} = 4\hat{i} + 2\hat{j} - \hat{k}$:

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 3 \\ 4 & 2 & -1 \end{vmatrix} = \hat{i}((-1)(-1)-(3)(2)) - \hat{j}((2)(-1)-(3)(4)) + \hat{k}((2)(2)-(-1)(4))$$

$$= \hat{i}(1-6) - \hat{j}(-2-12) + \hat{k}(4+4) = -5\hat{i} + 14\hat{j} + 8\hat{k}$$

SOLUTION 23

Self-Cross Product Properties

For any vector $\vec{V}$, $\vec{V}\times\vec{V} = 0$ because the angle between a vector and itself is $0°$, and $\sin(0°) = 0$. Component-wise, terms like $V_yV_z - V_zV_y = 0$ for all components.

SOLUTION 24

Torque

Given $\vec{r} = 4\hat{i} + 3\hat{j} + 5.5\hat{k}$ and $\vec{F} = -6\hat{i} + 12\hat{j}$:

(a) In unit vector notation:

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 3 & 5.5 \\ -6 & 12 & 0 \end{vmatrix} = \hat{i}(0-66) - \hat{j}(0-(-33)) + \hat{k}(48-(-18)) = -66\hat{i} - 33\hat{j} + 66\hat{k}\ \text{N}\cdot\text{m}$$

(b) Magnitude:

$$|\vec{\tau}| = \sqrt{(-66)^2 + (-33)^2 + 66^2} = \sqrt{4356 + 1089 + 4356} = \sqrt{9801} = 99 \text{ N} \cdot \text{m}$$

SOLUTION 25

Parallelogram Area

Given $\vec{a} = 3\hat{i} + 2\hat{j}$ and $\vec{b} = -\hat{i} + 4\hat{j}$:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 0 \\ -1 & 4 & 0 \end{vmatrix} = \hat{k}((3)(4) - (2)(-1)) = 14\hat{k}$$

Area $= |\vec{a} \times \vec{b}| = 14$ square units.

SOLUTION 26

Unit Normal Vector to a Plane

Compute cross product of $\vec{u} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{v} = \hat{i} + 3\hat{j} + \hat{k}$:

$$\vec{n} = \vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 3 & 1 \end{vmatrix} = 7\hat{i} - 4\hat{j} + 5\hat{k}$$

Normalize $\vec{n}$:

$$|\vec{n}| = \sqrt{7^2 + (-4)^2 + 5^2} = \sqrt{49 + 16 + 25} = \sqrt{90} = 3\sqrt{10}$$

Unit vector: $\hat{n} = \frac{7\hat{i} - 4\hat{j} + 5\hat{k}}{3\sqrt{10}}$.

SOLUTION 27

Three-Variable Cross Product Relation

Given $\vec{A} = 2\hat{i} + y\hat{j}$, $\vec{B} = 3\hat{j} + z\hat{k}$, and $\vec{C} = 12\hat{i} - 4\hat{j} + 6\hat{k}$:

(a) Compute $\vec{A} \times \vec{B} = (yz)\hat{i} - (2z)\hat{j} + (6)\hat{k}$. Equate to $\vec{C}$:

$$yz = 12, \quad -2z = -4 \implies z = 2, \quad 6 = 6$$

Using $z = 2$ in $yz = 12$ gives $2y = 12 \implies y = 6$.

(b) Verification: $\vec{C} \cdot \vec{A} = (12)(2) + (-4)(6) + (6)(0) = 24 - 24 = 0$ (Orthogonal). $\vec{C} \cdot \vec{B} =$

$(12)(0) + (-4)(3) + (6)(2) = -12 + 12 = 0$ (Orthogonal).

(c) Area of triangle $= \frac{1}{2}|\vec{C}| = \frac{1}{2}\sqrt{12^2 + (-4)^2 + 6^2} = \frac{1}{2}\sqrt{144 + 16 + 36} = \frac{1}{2}\sqrt{196} = \frac{14}{2} =$ 7 square units.

(d) Sine of angle: $|\vec{A} \times \vec{B}| = |\vec{A}||\vec{B}|\sin(\theta) \implies |\vec{C}| = |\vec{A}||\vec{B}|\sin(\theta)$.

$$|\vec{A}| = \sqrt{4 + 36} = \sqrt{40}, \quad |\vec{B}| = \sqrt{9 + 4} = \sqrt{13} \implies \sin(\theta) = \frac{14}{\sqrt{40}\sqrt{13}} = \frac{14}{\sqrt{520}} \approx 0.614$$